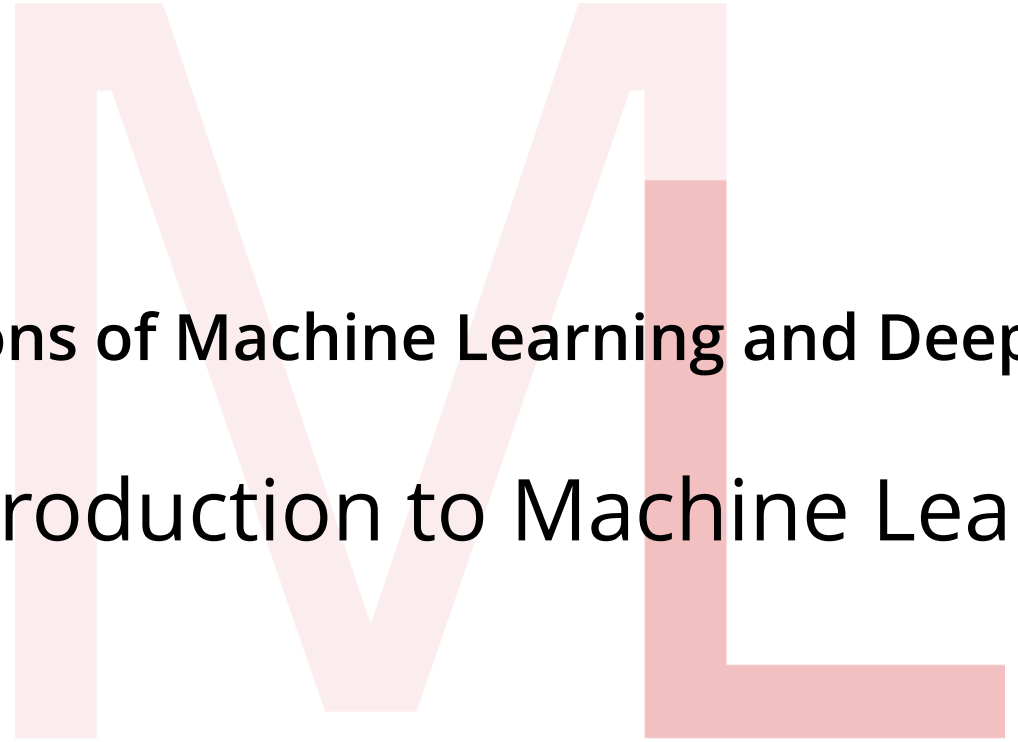
Foundations of Machine Learning and Deep Learning

1. Introduction to Machine Learning



upna
Universidad Pública de Navarra
Nafarroako Unibertsitate Publikoa

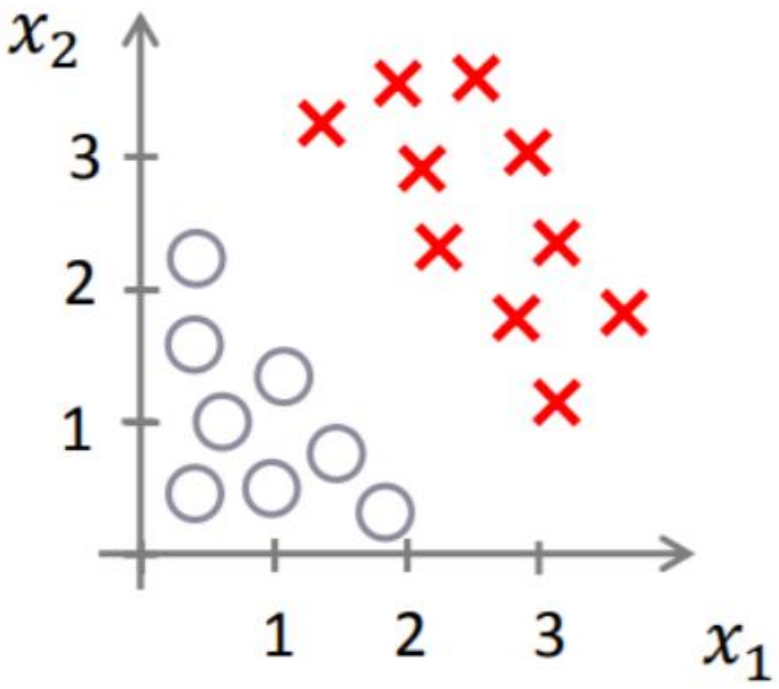
Outline

- What is Machine Learning?
- Notation & Core Concepts
- Learning paradigms
- Classical Models
  - K nearest neighbors
  - Naive Bayes
  - Decision trees
  - Ensemble models
- Model evaluation and selection
- Clustering

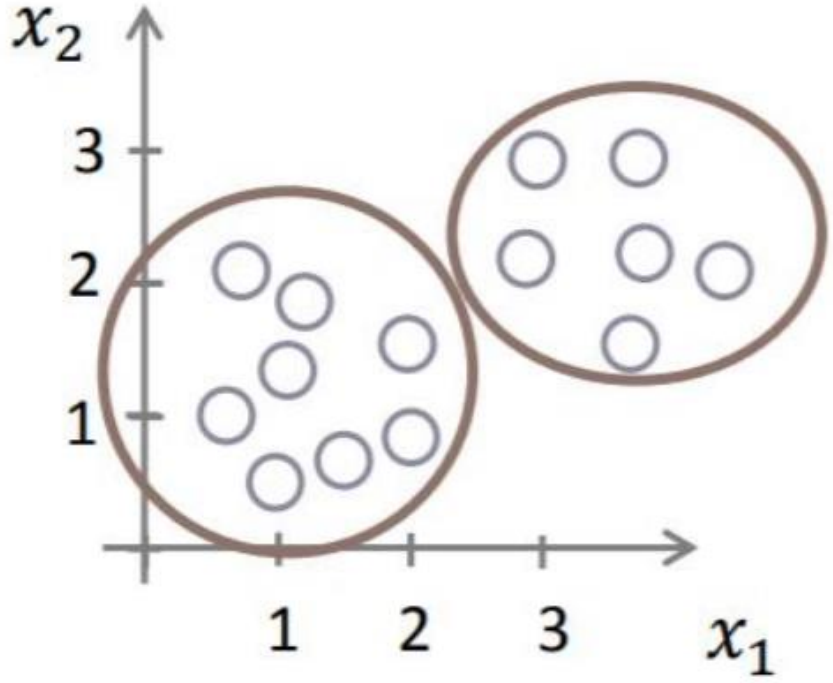
Clustering

- **What is clustering?**

Supervised



Unsupervised



Clustering

- **What is clustering?**
  - Unsupervised learning: grouping data without labels
  - Goal: to find natural structure in data
  - Objects in the same cluster should be more similar to each other than to objects in other clusters

- **Applications**
  - Customer segmentation in marketing
  - Document/topic grouping in NLP
  - Image compression
  - Anomaly detection



Clustering

- **What is clustering?**
  - Given a sample X of individuals, $\boldsymbol{x_1}, \ldots, \boldsymbol{x_m}$, each represented by an n-dimensional vector $\boldsymbol{x_j} = (x_{j1}, x_{j2}, \ldots, x_{jn}),\ j = 1, \ldots, m$, we must find a partition of the sample into regions $\omega_1, \ldots, \omega_c$ such that:
    - $\bigcup_{i=1}^{c} \omega_i = X$
    - $\omega_i \cap \omega_j = \emptyset; \quad i \neq j$
  - Special cases: soft/fuzzy clustering: clusters are not disjoint, data have membership degrees

Clustering

Core ingredients of clustering algorithms

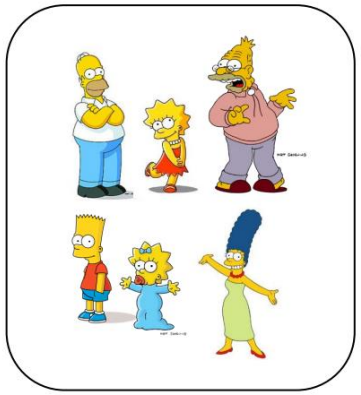
- Similarity measure (distance function)
  - Euclidean, Manhattan, cosine, etc.
- Objective function
  - E.g., minimize within-cluster variance
- Clustering strategy
  - Partitional (e.g., k-means)
  - Hierarchical (e.g., agglomerative)
  - Others (density based…)

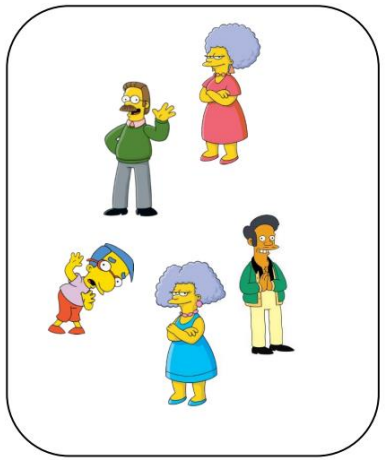
Similarity measure

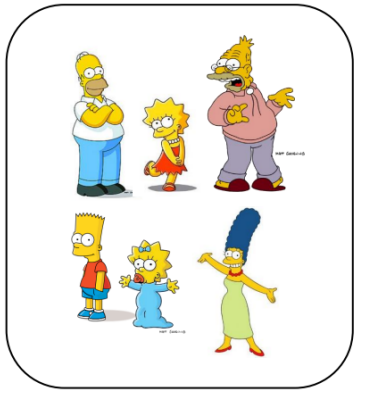
- What are the clusters in these data?



Smilarity measure

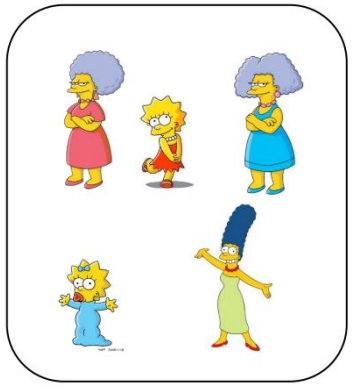




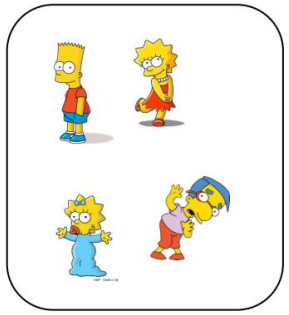


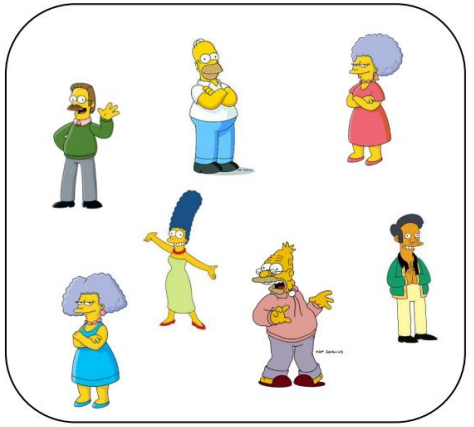








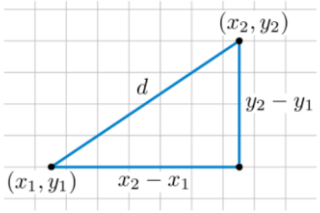


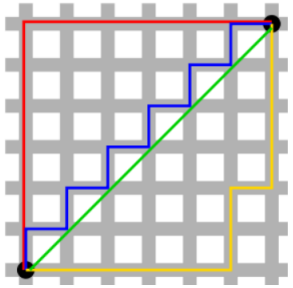


Similarity measure

- Numerical attributes (continuos)
  - Euclidean distance
  - Manhattan distance
  - Minkowski distance
  - Cosine distance
- Categorical attributes
  - Hamming distance
- Binary attributes
  - Jaccard distance
  - Dice distance

Similarity measures





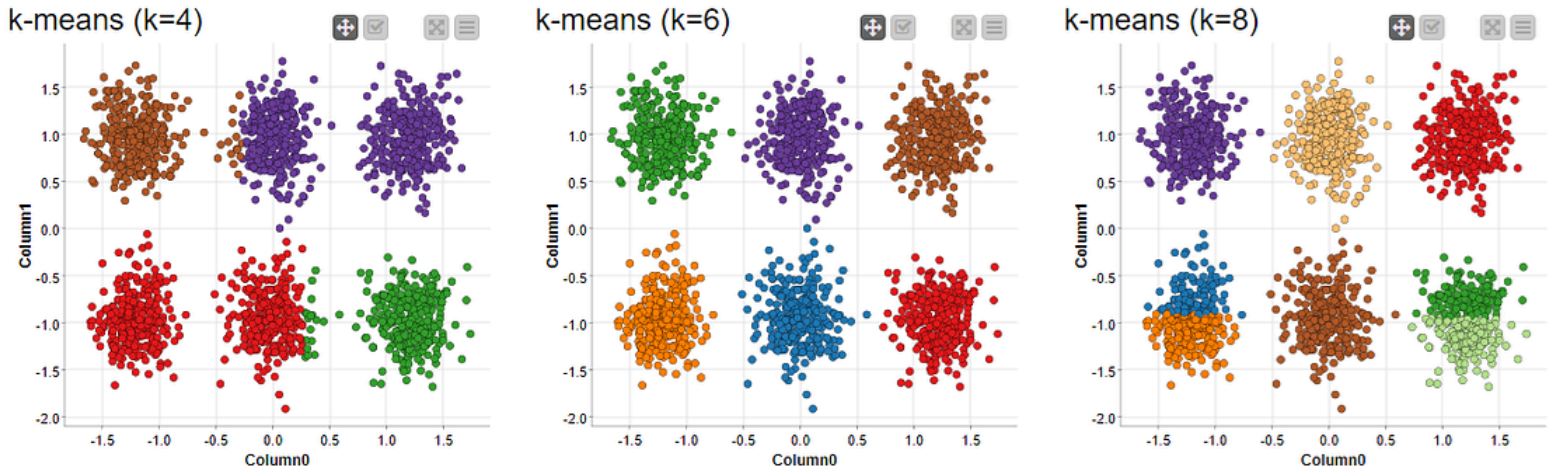
- Euclidean distance (L_2 norm, $||x||_2$)
  - Length of the line that connects two points
  - $d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$
- Manhattan distance (L_1 norm, $||x||_1$)
  - Sum of the projections' length
  - More robust to outliers in high dimension
  - $d(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n}|x_i - y_i|$
- Minkowski distance
  - Generalization of Euclidean ($p = 2$) and Manhattan ($p = 1$)
  - $d(\boldsymbol{x}, \boldsymbol{y}) = \left(\sum_{i=1}^{n}|x_i - y_i|^p\right)^{1/p}$
- Cosine distance
  - Measures the angle between vectors, ignoring their length
  - $d(\boldsymbol{x}, \boldsymbol{y}) = 1 - \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{||\boldsymbol{x}||_2 ||\boldsymbol{y}||_2}$

Similarity measures

- Hamming distance
  - Proportion of different attributes
  - $d(\boldsymbol{x}, \boldsymbol{y}) = \frac{\#\boldsymbol{different\ attributes}}{\boldsymbol{total\ attributes}}$
- Jaccard distance
  - Negative of intersection over union rate (do not consider examples that are false in both sets)
  - $d(\boldsymbol{x}, \boldsymbol{y}) = 1 - \frac{|\boldsymbol{x} \cap \boldsymbol{y}|}{|\boldsymbol{x} \cup \boldsymbol{y}|}$
- Dice distance
  - Negative of twice the intersection over the sum of the individual sises
  - $d(\boldsymbol{x}, \boldsymbol{y}) = 1 - \frac{2|\boldsymbol{x} \cap \boldsymbol{y}|}{|\boldsymbol{x}| + |\boldsymbol{y}|}$

Challenges in clustering

- No ground truth labels
- Different algorithms can give very different results
- Choosing the right number of clusters
- Sensitivity to:
  - Distance metric
  - Scaling of features
  - Outliers / noise



Clustering – k-means

K-means algorithm

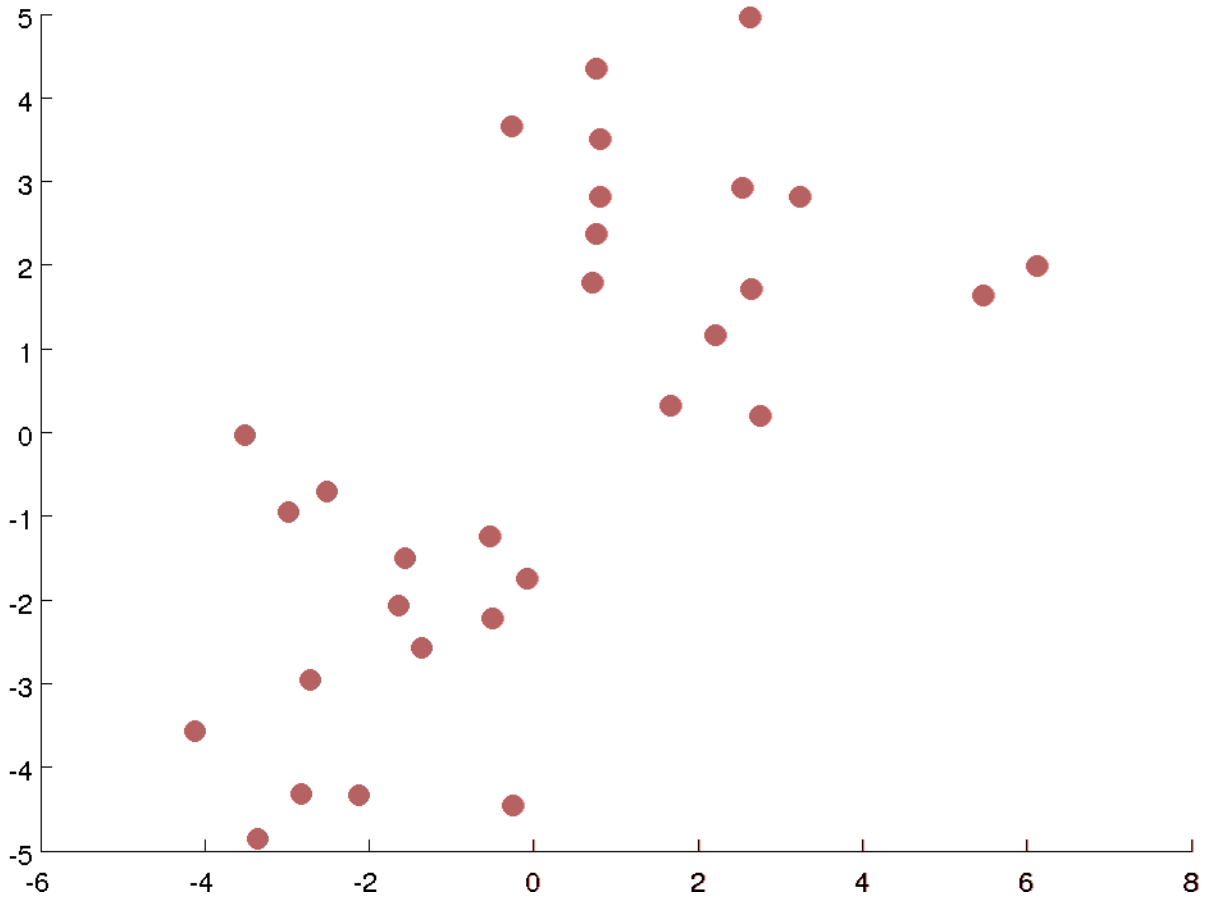
- Objective function

$$\min_{\{C_j\}} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

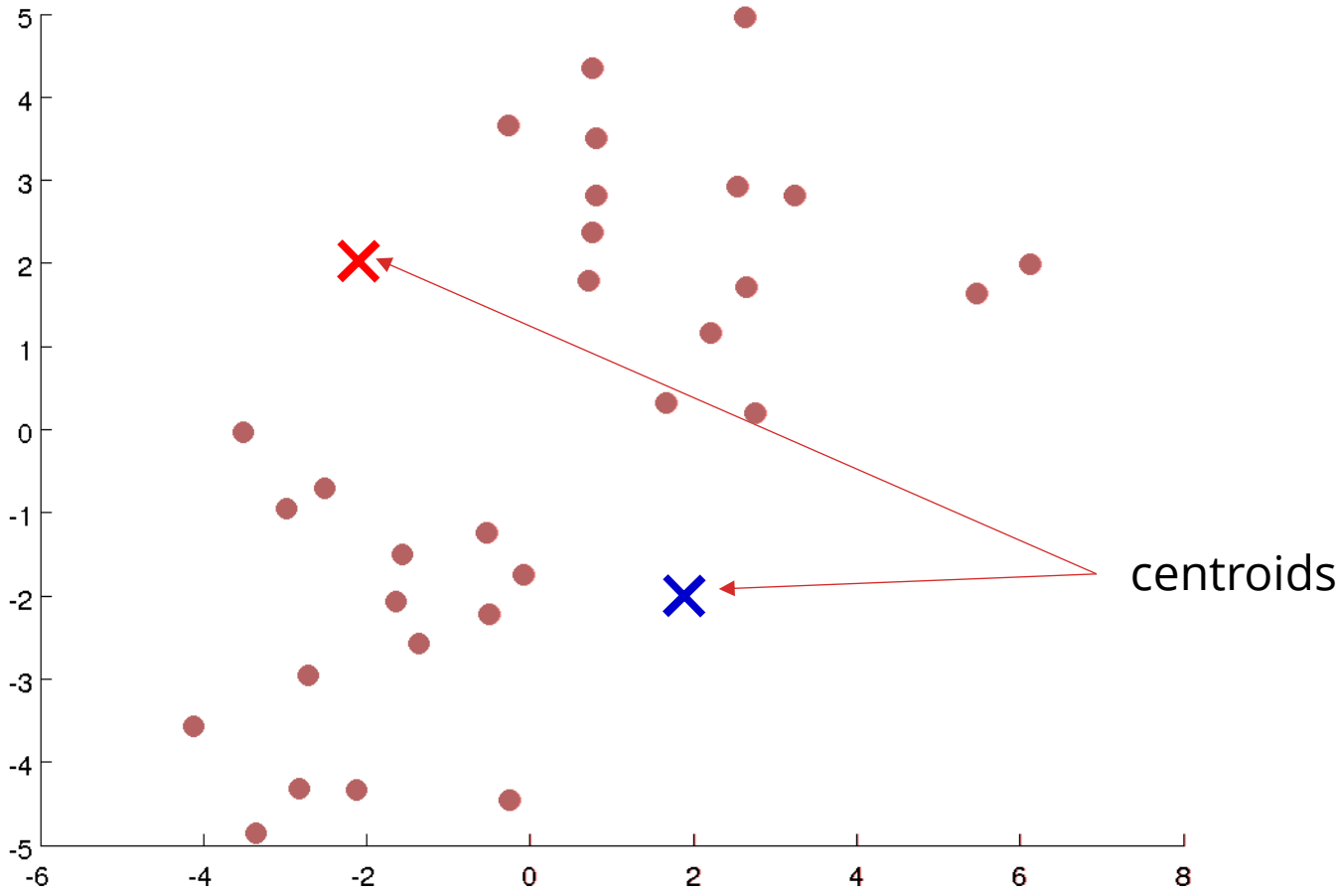
  - C_j: set of points in cluster j
  - μ_j: centroid (mean) of cluster j

- **Steps:**
  1. Choose *k* initial centroids (random)
  2. Assign step: assign each point to the nearest centroid
  3. Update step: recompute centroids as the mean of assigned points
  4. Repeat steps 2-3 until centroids stabilize or max iterations reached

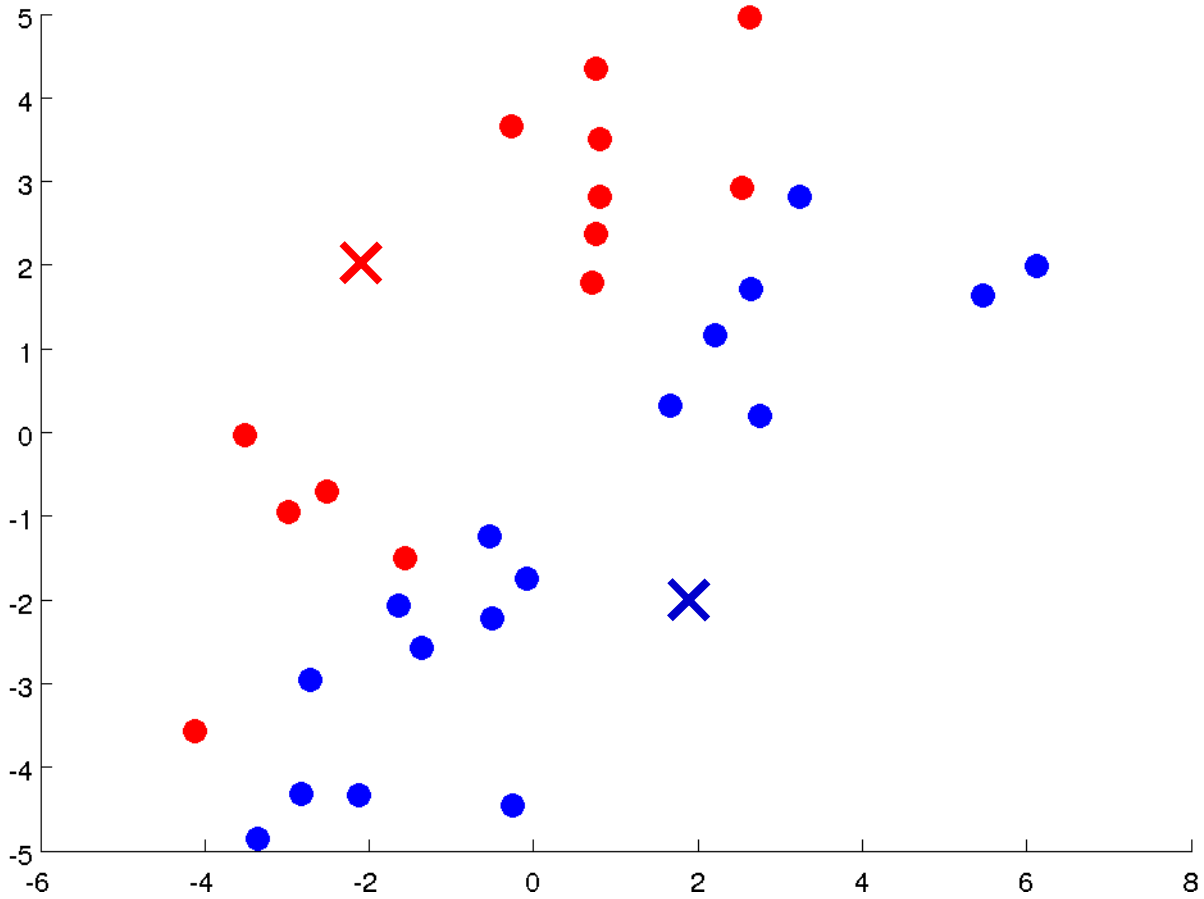
Clustering – k-means



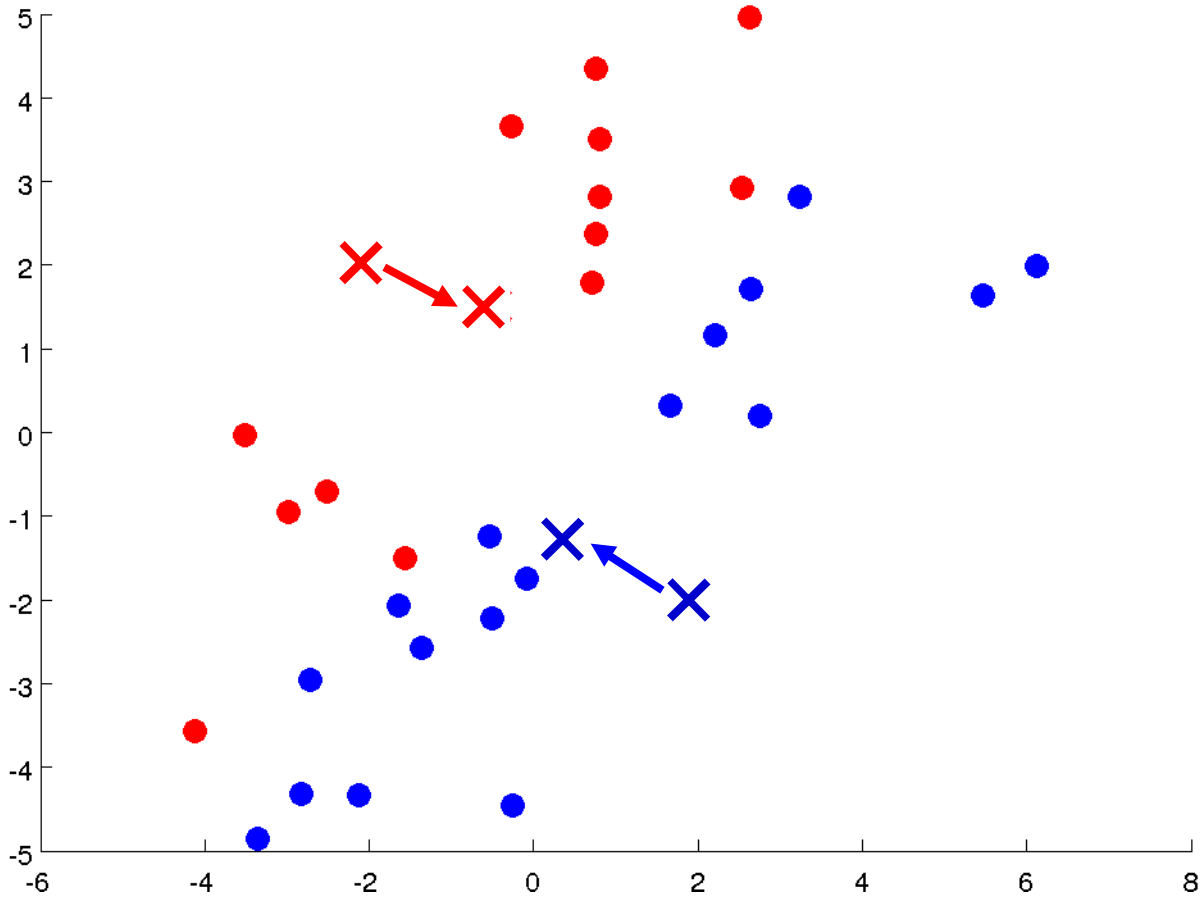
Clustering – k-means



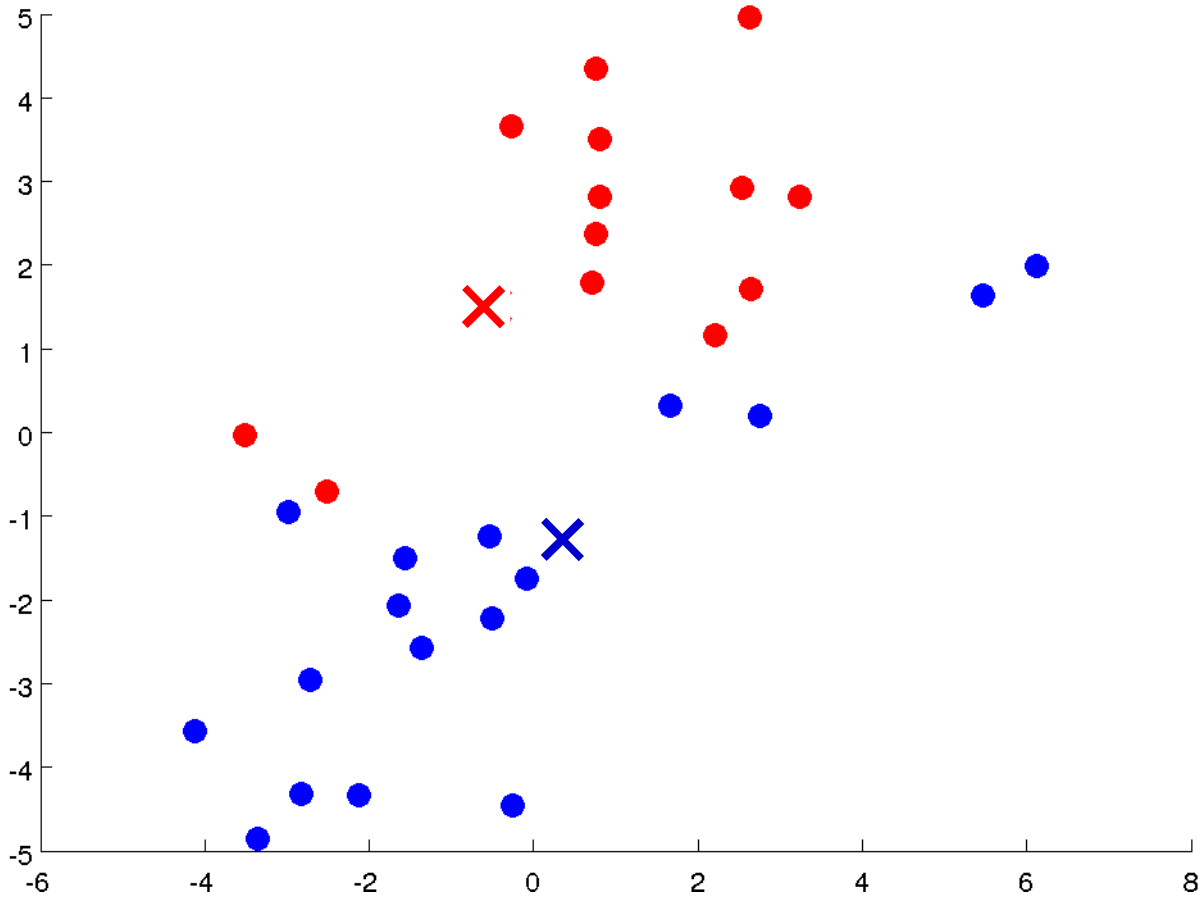
Clustering – k-means



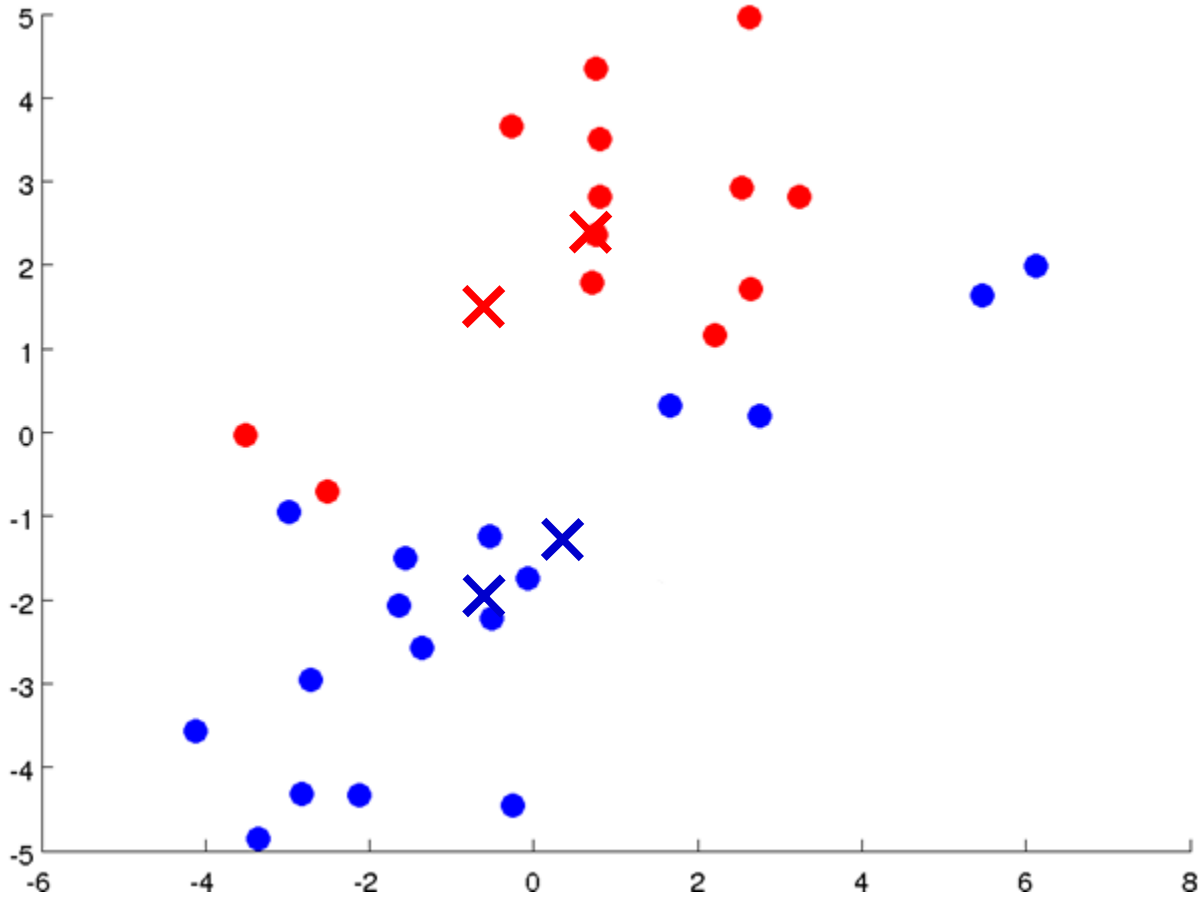
Clustering – k-means



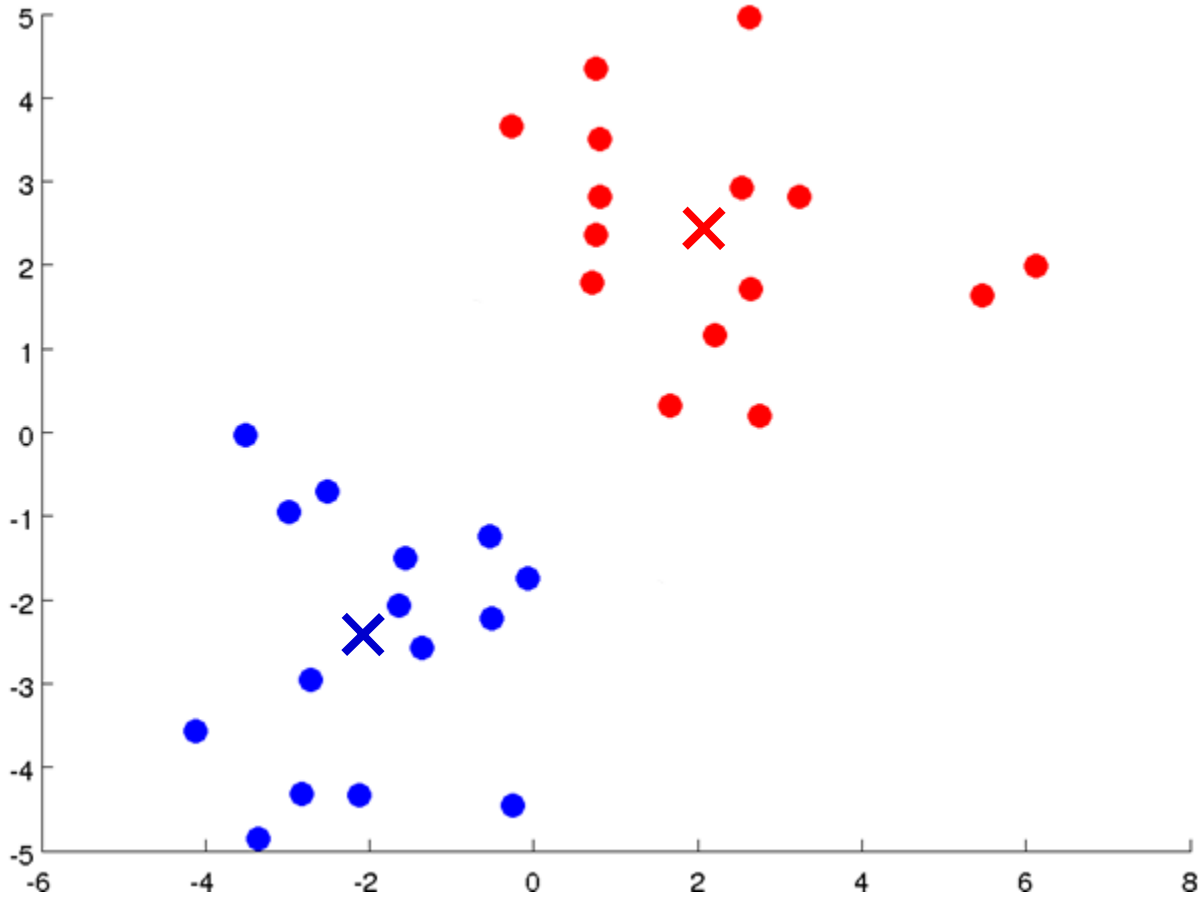
Clustering – k-means



Clustering – k-means



Clustering – k-means



Clustering – k-means

K-means algorithm

- Objective function

$$\min_{\{C_j\}} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

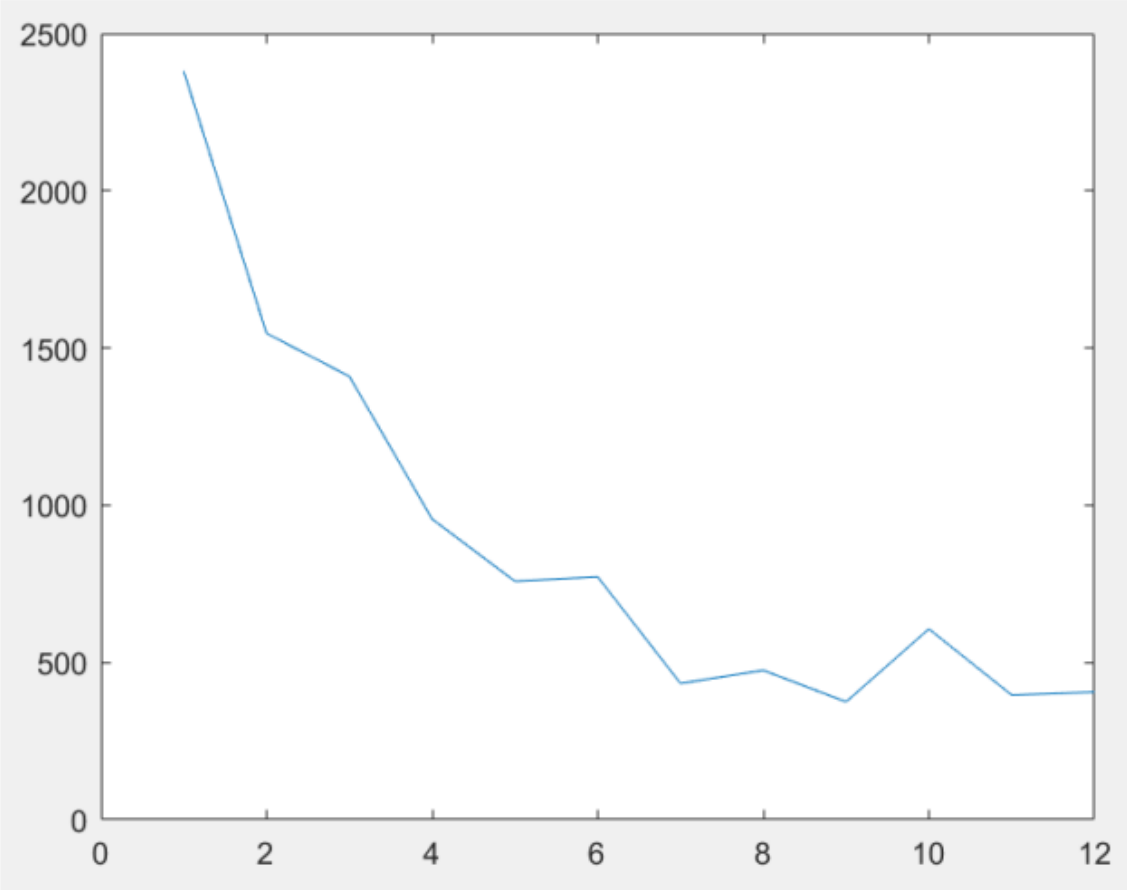
  - C_j: set of points in cluster j
  - μ_j: centroid (mean) of cluster j

- **Limitations:**
  1. Only considers spherical clusters
  2. Sensitive to initialization
  3. May stop al local minima
  4. Need to select the number of clusters

Use multiple runs

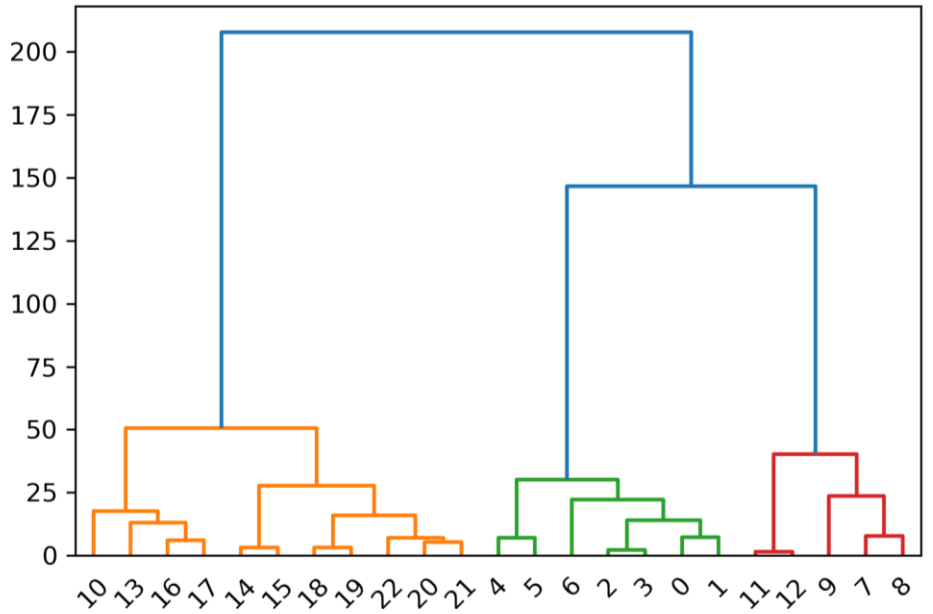
K-means algorithm

- Need to select the number of clusters → Elbow method



Clustering – Hierarchical clustering

- Agglomerative clustering
  - The goal is to build a hierarchy of clusters by successively merging the closest ones
- Divisive clustering
  - The goal is to build a hierarchy of clusters by successively splitting the most disimilar ones
- Both produce a dendogram: a tree of nested clusters
  - No need to pre-specify *k*, the tree can be "cut" at any level
  - Computationally more expensive than k-means



Clustering – Hierarchical clustering

Agglomerative clustering

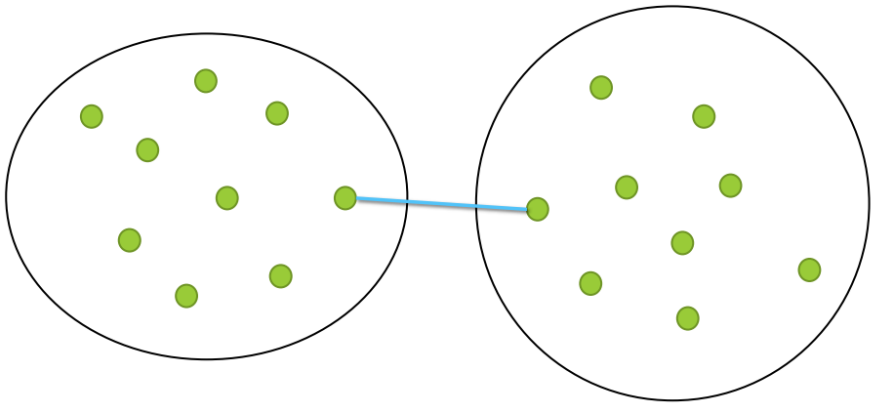
- **Steps:**
  1. Start with each data point as its own cluster
  2. At each step, merge the two most similar clusters
  3. Repeat until all points are merged into a single cluster

- Cluster similarity (linkage criteria)
  - Single linkage: min distance between members
  - Complete linkage: max distance between members
  - Average linkage: mean distance between members

Single linkage

- Minimum distance between a datum from cluster 1 and a datum from cluster 2

0	1	2	3	4	5
3	1	7	2	1	4
6	2	6	9	1	0



	0	1	2	3	4	5
0	0	4.47	4	3.16	5.39	6.08
1		0	7.21	7.07	1	3.6
2			0	5.83	7.81	6.71
3				0	8.06	9.22
4					0	3.16
5						0

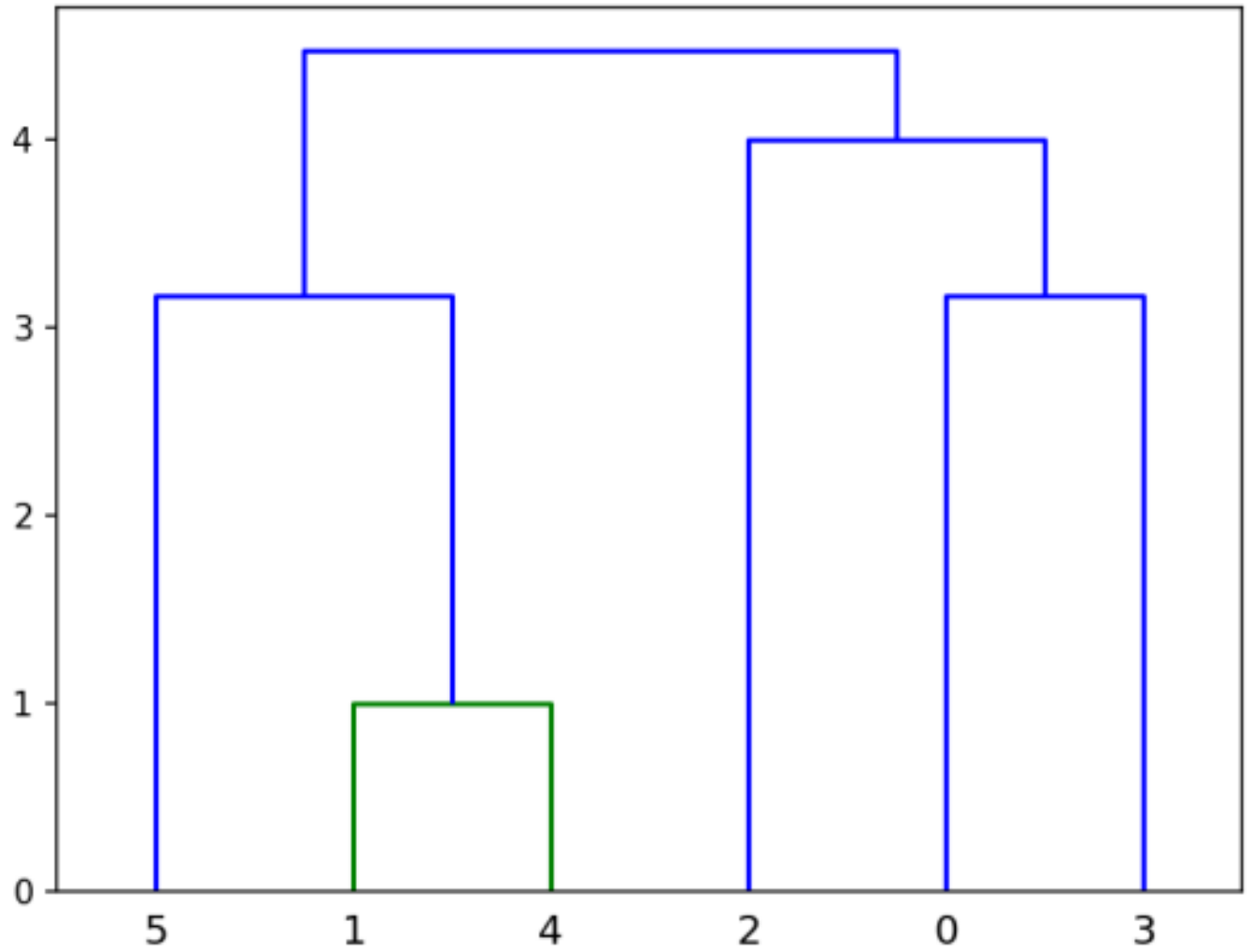
	0	2	3	5	6 (1-4)
0	0	4	3.16	6.08	4.47
2		0	5.83	6.71	7.21
3			0	9.22	7.07
5				0	3.16
6 (1-4)					0

	2	5	6 (1-4)	7 (0-3)
2	0	6.71	7.21	4
5		0	3.16	6.08
6 (1-4)			0	4.47
7 (0-3)				0

	2	7 (0-3)	8 (5-6)
2	0	4	6.71
7 (0-3)		0	4.47
8 (5-6)			0

	8 (5-6)	9 (2-7)
8 (5-6)	0	4.47
9 (2-7)		0

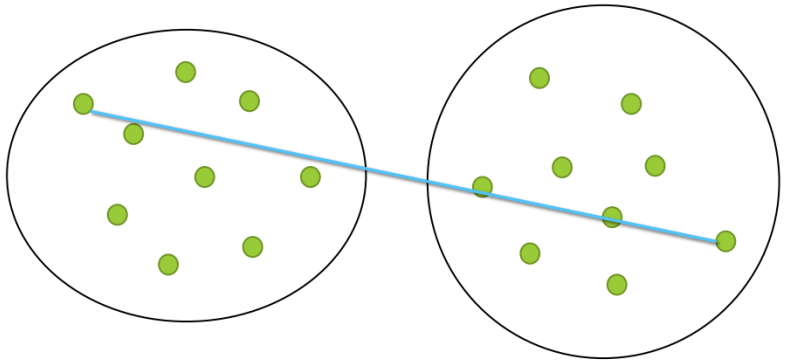
Clustering – Hierarchical clustering



Complete linkage

- Maximum distance between a datum from cluster 1 and a datum from cluster 2

0	1	2	3	4	5
3	1	7	2	1	4
6	2	6	9	1	0



	0	1	2	3	4	5
0	0	4.47	4	3.16	5.39	6.08
1		0	7.21	7.07	**1**	3.6
2			0	5.83	7.81	6.71
3				0	8.06	9.22
4					0	3.16
5						0

	0	2	3	5	6 (1-4)
0	0	4	**3.16**	6.08	5.39
2		0	5.83	6.71	7.81
3			0	9.22	8.06
5				0	3.6
6 (1-4)					0

	2	5	6 (1-4)	7 (0-3)
2	0	6.71	7.81	5.83
5		0	**3.6**	9.22
6 (1-4)			0	8.06
7 (0-3)				0

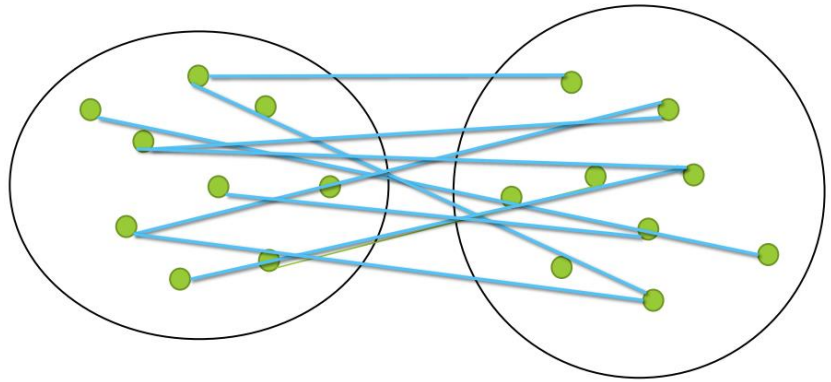
	2	7 (0-3)	8 (5-6)
2	0	**5.83**	7.21
7 (0-3)		0	9.22
8 (5-6)			0

	8 (5-6)	9 (2-7)
8 (5-6)	0	**9.22**
9 (2-7)		0

Average linkage

- Average distance between every pair of a datum from cluster 1 and a datum from cluster 2

0	1	2	3	4	5
3	1	7	2	1	4
6	2	6	9	1	0



	0	1	2	3	4	5
0	0	4.47	4	3.16	5.39	6.08
1		0	7.21	7.07	1	3.6
2			0	5.83	7.81	6.71
3				0	8.06	9.22
4					0	3.16
5						0

	0	2	3	5	6 (1-4)
0	0	4	3.16	6.08	4.93
2		0	5.83	6.71	7.51
3			0	9.22	7.57
5				0	3.38
6 (1-4)					0

	2	5	6 (1-4)	7 (0-3)
2	0	6.71	7.51	4.92
5		0	3.38	7.65
6 (1-4)			0	6.25
7 (0-3)				0

	2	7 (0-3)	8 (5-6)
2	0	4.92	7.25
7 (0-3)		0	6.72
8 (5-6)			0

	8 (5-6)	9 (2-7)
8 (5-6)	0	6.90
9 (2-7)		0

Clustering – Evaluation

Clustering performance metrics

- No labels available, therefore evaluation is harder
- No universal “best” metrics, the choice depends on data and application
- Metrics fall into two groups:
  - Internal indices: they rely only on the data and the clustering result
    - Silhouette score
    - Davies-Bouldin Index
    - Calinski-Harabasz Index
  - External indices: require ground-truth labels (for benchmarking)
    - Adjusted Rand Index (ARI)
    - Normalized Mutual Information (NMI)